Chamhembe

FLEET RENTAL PLATFORM

BUSINESS OVERVIEW · 15 APRIL 2026

A fleet-first rental platform built for scale.

A complete booking, deposit, inspection and billing system for car rental operators. Single-tenant today. Architected to go multi-tenant SaaS.

Prepared by · Chamhembe Engineering

Document · Platform Overview v1.0

Date · 15 April 2026

Audience · Operators, investors, technology partners

01 · EXECUTIVE SUMMARY

A fleet platform, not just a booking form.

Chamhembe is a full-stack car rental management platform built from the ground up around the operational reality of running a fleet — not a generic booking widget bolted onto a spreadsheet. Every lifecycle step a rental business actually does is a first-class citizen: commercial tiering, deposits, pickup and return inspections, mileage reconciliation, fuel policy, damage handling, invoicing, refunds, customer ratings, compliance tracking, and a mobile app for self-service.

The platform currently runs as a single-operator deployment (Chamhembe's own rental fleet in Harare), but the data model, permission system, and API surface are deliberately structured so that a multi-tenant SaaS rollout is a schema change away, not a rewrite.

30+	5	40+	99
API ENDPOINTS	USER ROLES	PERMISSIONS	AUTOMATED TESTS

The opportunity

Most rental operators in this market run on Excel, WhatsApp threads, and a phone tree. The ones that do use software pay for generic tools built for European or American workflows with payment rails (Stripe, PayPal) that don't work locally — no EcoCash, no OneMoney, no ZWL/USD split, no local compliance tracking. Chamhembe is the opposite: built in-market, built for the operational tempo of a Harare-sized fleet, and extensible.

02 · THE PROBLEM

Six things every rental operator gets wrong.

- **Deposit reconciliation is manual and error-prone.** Mileage overage, fuel shortfall, damage — most shops do it with a calculator and a pen, refund too much, and eat the difference every month.
- **Booking categories aren't a first-class concept.** Commercial terms (deposit, km allowance, excess rate, fuel policy) usually live in the operator's head or in a printed rate card. Customers get inconsistent quotes; staff waste time negotiating.
- **Vehicle lifecycle data is fragmented.** Fuel receipts in a drawer, service logs in a mechanic's WhatsApp chat, licence expiry dates on a wall calendar. No single source of truth means no profitability per vehicle.
- **Customer risk is invisible.** No rating history, no blacklist, no licence expiry check — so bad customers rebook after a fight and good customers get the same deposit as first-timers.
- **Pickup and return inspections are a liability.** Handwritten checklists disappear. Photos live on a staff member's personal phone. Disputes go unresolved because evidence is lost.
- **Customers can't self-serve.** Every enquiry is a phone call. Every booking confirmation is a WhatsApp forward. The operator becomes the bottleneck.

> **Our thesis:** none of these are exotic problems. They're just problems nobody built software for in this market. Chamhembe does.

03 · THE SOLUTION

What the platform actually does today.

Every feature below is shipping, tested, and running in production. Items marked ROADMAP are planned extensions.

Booking Categories

Commercial tiers (Small Car, Medium Sedan, SUV/Bakkie, Premium/Luxury) own the deposit, km allowance, excess rate, and fuel policy. Assign a vehicle to a tier; every booking on that vehicle uses those terms automatically.

Deposit Reconciliation

On return, the server deducts mileage overage, fuel shortfall, and damage from the deposit automatically and creates a refund payment for the surplus. No calculator. No disputes.

Pickup / Return Inspections

Structured checklists with photos, fuel level, odometer, and customer signature. Stored against the booking, diffable between pickup and return for damage claims.

Fuel Policy

Track pickup and return fuel level in quarter-tank units. Charge customers the category's per-level rate for any shortfall. Surplus fuel is never credited back — protects the operator from abuse.

Customer Profiles & Ratings

Every customer carries a licence number, expiry, emergency contact, and a five-dimension rating from past bookings (condition, timeliness, payment, communication, care of vehicle).

Blacklist & Greylist

One click to flag a customer. The booking creation flow refuses to serve blacklisted customers with a clear reason. Greylist lets you keep serving but with a warning.

Mobile Self-Booking

Native customer app: browse availability (no login), register, book, pay deposit, track booking status, view deposit refund after return. Full API-first design.

Fleet Admin Dashboard

Inertia React SPA for operators: vehicles, bookings, invoices, compliance, maintenance log, service providers, finance dashboard, and full reporting — all sub-100ms on Herd.

Compliance & Licences

Track ZINARA, ZBC, insurance, fitness, and custom licences per vehicle. Automatic expiry warnings before a vehicle rolls out non-compliant.

Maintenance Log

Record services, breakdowns, accidents. Link to service providers (mechanic, tow, panelbeater, parts, insurance). Each entry carries labour + parts + tow cost plus downtime days for PnL.

Vehicle Costs / Errands

Log fuel, parking, tolls, fines, car wash — anything the vehicle incurs outside a rental. Recording an odometer on an errand auto-bumps the vehicle's current reading.

Outsourced Fleet

Built-in support for third-party owned vehicles. Track the owner, payout rate, and Chamhembe markup. Each booking computes both the customer price and the owner payout.

Local Payment Methods

Cash, EcoCash, OneMoney, bank transfer, card, and customer wallet. Paynow gateway integration hooks for automated mobile money confirmations.

Invoicing & Receipts

Every completed booking auto-generates an invoice with a downloadable PDF. Every payment emits a printable receipt. Partial payments roll the invoice from sent → partially_paid → paid.

OpenAPI / Scalar Docs

Every API endpoint is auto-documented. A live Scalar UI lets mobile developers explore, authenticate, and test the API against any environment.

Multi-Tenant SaaS ROADMAP

Tenant-scoped data model, subdomain routing, tenant-level billing, branded customer app per tenant. Architected today for a small change, not a rewrite.

04 · HOW IT WORKS

The booking lifecycle end-to-end.

Every rental on the platform moves through the same state machine, with real money math at two specific points (creation and completion). No ad-hoc "how much should we charge?" conversations.

STEP	STATUS	WHAT HAPPENS
1 **Browse** Customer or staff	n/a	Customer (via mobile app) or staff (via admin) searches availability. Vehicles show daily rate + category terms (deposit, km/day, excess rate, fuel charge). Unavailable vehicles are excluded.
2 **Create** Booking created	`pending`	Server snapshots `daily_rate`, `km_allowance`, `excess_km_rate`, and `deposit_amount` from the vehicle's category. Computes `base_amount = rental_days × daily_rate`. Customer sees **hire fee + deposit = total due now**.
3 **Confirm** Staff approves	`confirmed`	Booking agent reviews and confirms. Vehicle is now reserved for those dates — no one else can book it. Customer gets a confirmation notification.
4 **Deposit captured** Payment recorded	`confirmed`	Deposit and (optionally) first-day hire fee captured via EcoCash, cash, or bank transfer. Linked to the booking. Invoice generated and marked `sent`.
5 **Activate** Customer picks up	`active`	Staff records pickup odometer + fuel level, captures pickup inspection (photos, checklist, customer signature). Vehicle status flips to `rented`.
6 **Complete** Customer returns	`completed`	Staff records return odometer + fuel level + return inspection. Server computes mileage overage + fuel charge + damage charge, deducts from deposit, creates a `deposit_refund` payment for any surplus, or adds the shortfall to the invoice.
7 **Settle** Invoice closed	`paid`	Any invoice balance is paid off; invoice rolls to `paid`. Customer leaves a rating on the rental experience; staff leaves a rating on the customer (reputation for next time).

> **Worked example.** 3-day Honda Fit rental (Small Car tier): daily rate $45, 200 km/day allowance, $0.50/excess km, $15 per quarter-tank short, $150 deposit. Customer pays **$135 hire + $150 deposit = $285 on pickup**. Drives 800 km and returns at half tank. Overage: (800 − 600) × $0.50 = $100. Fuel: 2 levels short × $15 = $30. Deductions: $130. Refund: $150 − $130 = **$20 auto-refunded**. Final customer cost: $265. Zero negotiation. Zero calculator time.

05 · MULTI-TENANT FUTURE

Why SaaS is the next chapter.

Chamhembe is currently deployed as a single-tenant installation serving one operator: Chamhembe Rental Services. The platform was built that way on purpose — prove the model end-to-end for one real business before generalising. The proof is in the platform you're reading about.

The next phase is multi-tenant SaaS: one Chamhembe codebase hosting multiple independent rental operators, each with their own fleet, customers, staff, branding, and billing.

What multi-tenant unlocks for operators

- **Sign up in 10 minutes.** Pick a subdomain, add your first vehicle, go live. No install, no sysadmin, no dev team.
- **Pay only for what you use.** Per-vehicle or per-booking pricing tiers instead of six-figure software licenses.
- **Branded mobile app, zero app-store work.** Customers see your logo and colours. Chamhembe ships updates to everyone.
- **Full data isolation.** Row-level tenant scoping — no accidental data leakage between operators.
- **Shared learnings, shared upgrades.** Every feature Chamhembe ships lands on day one for every operator on the platform.

What's already in place to make this a short path

- **Clean permission system.** Roles (super-admin, fleet-manager, booking-agent, finance, customer) and fine-grained permissions are already tenant-ready — add a tenant_id scope and role inheritance follows.
- **Domain-driven services.** Business logic lives in `BookingService`, `PricingService`, `InvoiceService`, `PaymentService`. No controller has a calculator in it, which means adding a tenant layer doesn't break the math.
- **API-first everything.** The mobile app and the admin dashboard both talk to the same v1 REST API. A tenant-branded mobile app is a config swap, not a rewrite.
- **Tested pricing engine.** 99 automated tests protect the money math. A tenant-scoping refactor stays safe because the tests tell you immediately if anything moves.

06 · PRICING MODEL

Tiered pricing for multi-tenant rollout.

Indicative SaaS pricing once multi-tenant rollout is complete. Everything is in USD and subject to local tax. All tiers include the mobile customer app, unlimited bookings, unlimited staff accounts, and daily backups.

STARTER

$49 / month

For new operators with up to 10 vehicles.

- Up to 10 vehicles
- Unlimited bookings
- 3 staff accounts
- Email support
- Shared mobile app

PROFESSIONAL

$149 / month

For growing fleets — the sweet spot.

- Up to 50 vehicles
- Unlimited bookings
- 10 staff accounts
- Priority support
- Branded mobile app
- Advanced reports
- API access

FLEET

$499 / month

For multi-location operators with 50+ vehicles.

- Up to 200 vehicles
- Unlimited staff
- Dedicated support
- White-labelled mobile app
- Custom report builder
- Webhooks

ENTERPRISE

Contact / custom

For 200+ vehicles, on-prem, or bespoke integrations.

- Unlimited everything
- SLA + 24/7 support
- On-prem or dedicated cloud
- Custom feature development
- Single sign-on

Unit economics for the operator

Most rental operators lose **$150–$400 per month** on deposit reconciliation errors alone — refunding too much, forgetting to charge mileage overage, absorbing fuel shortfalls. The Professional tier pays for itself on **the first two bookings it reconciles correctly**. Everything after that is pure margin recovery.

07 · TECHNOLOGY

Boring tech, fast iteration.

Chamhembe is built on mainstream, boring, well-documented technology chosen deliberately so any Laravel or React developer can pick up the codebase and be productive on day one. No custom framework, no proprietary runtime, no vendor lock-in.

BACKEND	Laravel 13 · PHP 8.4 · MySQL · Sanctum · Spatie Permission · DomPDF · Scramble OpenAPI · Scalar
ADMIN WEB	Inertia v3 · React 19 · TypeScript · Tailwind CSS 4 · shadcn/ui · Wayfinder · Vite
MOBILE	React Native (Expo) · TanStack Query · Zustand · EAS Build · OTA updates
QUALITY	99 automated tests · PHPStan · Pint · ESLint · TypeScript strict · GitHub Actions CI
PAYMENTS	Paynow (EcoCash, OneMoney) · Cash · Bank transfer · Card · Customer wallet

08 · WHY CHAMHEMBE

The case for building this, not buying it.

The global rental management software market is dominated by products built for North American and European operators: monthly lease rates in euros, Stripe-based payments, LTD/Ltd compliance workflows, and zero support for split-currency billing. These platforms cost thousands of dollars per month, require weeks of setup, and still don't let a customer pay a deposit on EcoCash.

Chamhembe is the first rental platform built *from* this market, *for* this market. It speaks ZWL and USD fluently, supports local mobile money out of the box, tracks ZINARA and insurance as first-class compliance items, and is run by a team that uses it for their own rental fleet every single day — meaning every friction, every edge case, every "oh we need this too" ends up in the backlog by the next sprint.

We use what we sell. Chamhembe Rental Services runs its entire fleet on this platform. Every deposit reconciled, every inspection logged, every invoice paid — runs through the same code we'd ship to another operator tomorrow. That's the best product feedback loop in software.

09 · WHAT'S NEXT

Let's talk.

Three ways to engage, in increasing order of commitment:

- **See the live platform.** We'll walk you through the admin dashboard, the mobile app, and the booking → deposit reconciliation flow on real data. 30 minutes, no slides.
- **Pilot with your fleet.** Spin up a single-tenant deployment with your vehicles and customers. Run it in parallel with your current system for a month. No cost, no commitment.
- **Partner on the multi-tenant rollout.** Operators willing to be launch partners get founding-member pricing locked in for the first year and a direct line to the engineering roadmap.

Contact: Chamhembe Engineering

Email: `hello@chamhembe.co.zw`

Platform docs: `https://car-rental-web.test/docs/api`

Live API: `https://car-rental-web.test/api/v1`